

Mixed Reality Piano Instruction Framework

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Task Description

Mixed Reality (MR) applications for immersive musical instruction often focus on specific use cases, such as head-mounted display (HMD) visualization or desktop-based lesson displays. The goal of this project is to create a unified, modular, and extensible framework for MR-based piano instruction.

The task of this project is to analyze the current software landscape and integrate separate implementations into a structured architecture with separate modules. The framework should support modular authoring pipelines for piano lessons, initially based on MIDI files, while allowing future integration of additional formats or live input sources. Optionally, hand-tracking sensors may be integrated to simultaneously record and author synchronized virtual 3D hand animations alongside piano performances.

The system should further support interchangeable visualization modules for piano instruction. These may include colorful bar-based note guidance, automatically generated sheet music displays, and virtual hand animations derived from captured hand-tracking data. Different visualization methods should be combinable depending on the learner’s needs.

Finally, the framework should generalize both display and tracking components. Lessons should be deployable across immersive MR headsets (e.g. *Meta Quest 3S*), desktop monitors, and projectors without substantial redevelopment. Spatial alignment and piano tracking should also be modular, initially supporting marker-based approaches such as Vuforia tracking while remaining extensible toward markerless piano detection methods.

The application should be implemented in the Unity3D game engine and should emphasize clean software engineering practices, scalability, and ease of extension. Other functionality beyond the discussed points could be explored as well.